



MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH. (COMPUTER SCIENCE & ENGINEERING) DEGREE EXAMINATIONS - APRIL/ MAY 2025

SUBJECT: CSE 2223 / CSE_2253 EMBEDDED SYSTEMS EMBEDDED SYSTEMS [CSE 2223]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) Develop an ARM assembly program that counts the number of times a given 32-bit number appears in an array of ten numbers available in code memory. Store the count in R8. (5)
- 1B) Develop an ALP to find GCD of two 32-bit numbers. (3)
- 1C) Explain the LDM and STM instructions in ARM with an example. (2)
- 2A) Develop an ARM assembly language program that does the following operations in an empty ascending stack. Assume Stack_Pointer = 0xFF8.
i) Load the registers R0, R1 and R2 with 0x12A, 0x14B and 0x5D respectively.
ii) Call a subroutine that pushes the register content onto the stack using STM.
iii) Make the values of R0, R1, R2 as zero. (4)
iv) Restore the contents of the original register from the stack using LDM and return to the caller.
Add the sum of the three registers and store the result in register R3.
- 2B) Find the address space range of each of the following memory of an ARM chip
i) 4KB of EEPROM starting at address 0x90000000
ii) 8KB of SRAM starting at address 0xA0000000 (3)
iii) 16KB of Flash ROM starting at address 0xE0000000
- 2C) Develop an Embedded C program to display ODD hexadecimal numbers between 1 and 15 in the Seven-Segment Display (SSD) unit interfaced with LPC 1768. Assume that the eight segments of the SSD unit are connected to the port pins P0.4 to P0.11 and there is only one SSD unit which can be enabled using a value of 1. The enable line of SSD unit is connected to the pin P0.19. (3)
- 3A) With neat diagrams, explain the types of PWM signals that can be generated with LPC1768. Given PCLK = TCLK = 6Mhz, PR=0, obtain the values of MR0 and MR1 to generate a PWM signal of 20ms period and 65% duty cycle. (5)
- 3B) Develop an embedded C program using LPC1768, to generate square waveform. The Aout pin is P0.26 which should be configured with function 2 (second alternate function). (3)
- 3C) Write Embedded C statements to perform the following: (2)
i) Read the status of **P0.12 to P0.18** and ignore other bits.
ii) Store the value of **P1.5 to P1.10** in a variable after shifting them to the LSB position.

- 4A) Develop an embedded C program using LPC1768 Timer0 to generate a periodic interrupt every 2 milliseconds. Assume PCLK = 25 MHz and Tres = 1 millisecond. The generated interrupt should toggle LEDs connected to P2.10 to P2.13. (4)
- 4B) Explain the purpose and usage of Prescale Register (PR), Prescale Counter (PC) and Timer Counter (TC) in LPC1768 Timer Programming. (3)
- 4C) Construct an ISR (ADC_IRQHandler) to display the digital output of an analog signal in the LED unit. The analog signal is assumed to be a current signal that ranges from 0 to 15 A and is available at ADC channel 3. The digital output should be displayed in binary form in the LED unit. The LED unit is assumed to have 12 LEDs and the port pins P2.10 to 2.21 are connected to the LED unit. (3)
- 5A) Write Embedded C statements for configuring UART0 in LPC1768 with the following specifications:
- i) Configure P0.2 and P0.3 with function 1 as TX and RX pins for UART0.
 - ii) Length of the word: 6, no. of stop bits: 2, parity: even parity.
 - iii) Enable both Rx/Tx FIFO operations and reset both Rx and Tx FIFO.
 - iv) If DLEst = 355, configure Divisor Latch registers.
 - v) Configure the appropriate register if DIVADDVAL and MULVAL are 8 and 10 respectively.

The structure of FCR and LCR are as follows:

FCR

31:8	7:6	5:4	3	2	1	0
RESE RVED	RXTRI GGER	RESE RVED	DM A MO DE	TX FIF O RE SE T	RX FIF O RE SE T	FIFO ENA BLE

(5)

LCR

31:8	7	6	5:4	3	2	1:0
Reser ved	DL AB	Break Contro l	Parit y Selec t	Parity Enabl e	Stop Bit Select	Word Length Select

- 5B) Discuss the concept of Nested Vector Interrupt Controller (NVIC) in LPC1768. Explain how it ensures prioritization of interrupts. (3)
- 5C) Explain the steps to configure **External Interrupts (EINT0-EINT3)** on LPC1768. Discuss the roles of EXTMODE and EXTPOLAR registers in this configuration. (2)

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